Predicting Citi Bike Station Availability

Supporting Efficient and Equitable Operations

CUNY 622 | ML in Production | Revised project plan | October 1, 2026

The class's five-minute station snapshots make empty/full-station forecasting feasible. Predict when stations will run out of rentable bikes or usable docks, then use those predictions to support cost-conscious, equitable service decisions. Docking costs remain a decision-support extension around the deployed ML system.

What changes in the existing plan

Earlier plan	Revised plan
Forecast completed departures and arrivals	Predict empty/full risk from five-minute status histories.
Collect availability ourselves	Use the class collection; verify coverage and freshness.
Import the full trip archive first	Begin with the shared catalog; add selected trip history where useful.
Emphasize annual station investment	Emphasize operational decisions during the four-week production period.
Analyze costs against historical utilization	Combine cost scenarios with predicted problems and observed reliability.
Produce models and analysis	Deploy scheduled predictions, an application, and outcome monitoring.

First task: Audit the class tables for earliest dates, station coverage, missing intervals, schemas, and permissions. Do not assume they contain a complete year.

Recommended machine-learning problem

For each operating station, every 15 minutes, predict (1) the probability of no rentable bikes 30 minutes later and (2) the probability of no usable return docks 30 minutes later. These are two supervised binary-classification tasks. They can occur separately and should not be forced into mutually exclusive categories.

The MVP predicts conditions at the 30-minute horizon. Predicting whether a station becomes empty or full at any point during the next 30 minutes is an extension requiring complete future observation windows.

Distinguish shortages from disabled service. GBFS provides availability counts, reporting timestamps, and operating flags; confirm the feed version and class schema. Missing or stale observations should produce an unavailable label rather than a functioning-station assumption. [1-2]

Proposal requirements and model design

Requirement	Recommended answer
1. ML problem and use	Predict 30-minute empty/full risk. An operations dashboard identifies stations needing attention; a rider application could show likely availability.
2. Data sources	Class station snapshots, daily information, weather and trip data. Add ACS transportation indicators and tract boundaries as documented shared tables.
3. Initial models	Persistence baseline, logistic regression, and gradient-boosted trees. Separate empty and full classifiers.
4. Evaluation	Chronological backtesting; precision-recall AUC, Brier score, calibration, and precision/recall among highest-priority alerts. Repeat on production forecasts.
5. Training approach	Weekly batch retraining; inference every 15 minutes using latest available snapshots.
6. Compute	CPU-based tabular models with distributed feature preparation. No planned LLM, GPU, or large neural network.
7. MVP	Scheduled predictions written to a table, a risk dashboard, and an evaluation job comparing stored forecasts with later observations.

Features and validation

Useful features include current bike/dock counts, availability fractions, changes over the past 15-60 minutes, capacity, operating flags, hour, weekday, and weather available at prediction time. Historical trip aggregates are optional additions, not prerequisites.

Use chronological splits and ensure labels have matured before each cutoff. Random splits of neighboring snapshots would make evaluation overly optimistic.

Required additional dataset: ACS transportation access

ACS neighborhood indicators support the social-betterment objective. Census table B08201 contains household vehicle-availability information, available through the Census API. Use appropriate five-year estimates for tract analysis. [3-4]

Publish a documented tract-level table containing tract GEOID and geography vintage; total households and households without vehicles; percentage without vehicles; estimates and margins of error; source year, retrieval date, and field definitions. Publish a separate station-to-tract crosswalk.

These indicators describe neighborhood transportation need, not individual riders' circumstances. Initially use them for dashboard context and performance comparisons rather than prediction features.

Financial assessment and production system

Where costs and dock use fit

Assessment	What the analysis can establish
Station utilization	Trips per dock-day, availability patterns, and observed empty/full frequency.
Operational priority	Predicted service risk, recurrence, and neighborhood transportation need.
Financial scenarios	Estimated costs of additional docks, maintenance, or rebalancing under stated assumptions.

A forecast cannot establish how many trips adding docks would generate. Snapshot changes alone cannot distinguish rider activity from rebalancing and maintenance. Intervention benefits and savings remain scenario estimates unless stronger operational evidence becomes available.

For the MVP, show risk and access indicators side by side. Introduce a combined priority score only after documenting its weights and testing how they change rankings.

Production design

Use the existing class ingestion as the starting point:

Job / component	Function
1. Feature job	Validate freshness and create station features.
2. Prediction job	Every 15 minutes, store forecast origin, target time, probabilities, model version, and data-quality status.
3. Dashboard	Show current availability, predicted risk, access indicators, and prediction freshness.
4. Evaluation job	Match forecasts to subsequent valid observations and calculate daily performance.
5. Training job	Retrain weekly; compare candidates against existing models and baselines. Retain the previous model if acceptance checks fail.

Monitor prediction coverage, stale inputs, missed jobs, scoring time, calibration, and alert accuracy. Preserve predictions before outcomes occur so the final production evaluation is reproducible.

Core recommendation

Make availability forecasting the production project, ACS access analysis the required external-data contribution, and docking costs a bounded decision-support extension. This provides measurable outcomes during deployment and fits the assignment's emphasis on an end-to-end production system.

Course timeline and documentation

Deadline	Target
October 4	Maximum three-page proposal answering all seven questions; references separate.
October-early November	Audit data, publish external dataset, build labels/features, and establish baselines.
November 2-7	Demonstrate end-to-end MVP. Three slides: purpose, architecture, and remaining challenges.
Before November 16	Validate scheduling, evaluation, recovery, and dashboard freshness.
November 16-December 14	Run production predictions and monitoring for four weeks.
Week of December 14	Live demonstration and measured production performance.
December 18	Model card, external-data datasheet, runbook, and approximately 2,000-word technical report.

Documentation deliverables

Model card: Intended use, training data, evaluation procedure, production performance, limitations, and performance differences across relevant station/neighborhood groups. [5]

Datasheet: External-data motivation, composition, provenance, collection process, intended uses, field definitions, and limitations. [6]

Runbook: Jobs and schedules, model and table locations, logs, evaluation storage, failure recovery, and model replacement procedures.

Technical report: Problem definition, system design, ML components, application demonstration, and reflection on production results; target approximately 2,000 words.

References

1. GBFS: Get started

2. GBFS field definitions

3. Census ACS five-year table B08201 (example schema)

4. Census ACS API

5. Model Cards for Model Reporting

6. Datasheets for Datasets

Course requirements and deadlines are taken from the supplied assignment. This document is a planning brief; it is not the final three-page proposal. The example ACS schema link does not prescribe the vintage to use.